

Jing Yang

✉ yangj668@mail2.sysu.edu.cn

📞 / 🌐 (+86) 17501529298

Guangzhou, China
Institute of Logic and Cognition
Sun Yat-sen University (SYSU)

Education

B.A.	Sun Yat-sen University <i>Bachelor of Arts in Logic, Department of Philosophy</i> • GPA: 3.7/4.0 IELTS: 7.5 • Major Coursework: Principles of Artificial Intelligence (90), Data and Text Analysis (93), Probability and Rational Choice (95), Algebraic Structures (90), Proof Theory (90).	Sep. 2022 – Jun. 2026
------	---	-----------------------

Research Experience

Research Intern	HCP Lab, Sun Yat-Sen University <i>Supervisor: Dr. Keze Wang</i> • Conducted research on LLM-based agents and workflow optimization , focusing on improving reliability, robustness, and efficiency in tool-using and multi-step reasoning systems. • Developed methods for failure analysis, adaptive workflow refinement, and multi-agent coordination , including failure-mode clustering, graph-level workflow editing, and propose-and-verify optimization loops. • Explored efficient LLM training and inference mechanisms , including probabilistic belief updating, approximate-gradient optimization, and memory-constrained full-parameter fine-tuning.	Apr. 2024 – Present
-----------------	--	---------------------

Selected Publications

- Jing Yang, Jusheng Zhang, & Keze Wang. NBCG: Nash-Bargained Causal Game for Long-Tailed Multi-Label NLP. ICML, 2026. (Accepted)** ICML 2026
- Jing Yang, Yijia Fan, Jusheng Zhang, & Keze Wang. Agent-GSPO: Communication-Efficient Multi-Agent Systems via Group Sequence Policy Optimization. ICASSP, 2026. (Accepted)** ICASSP 2026
- Jusheng Zhang, Jing Yang, Kaitong Cai, Ziliang Chen, Yongsen Zheng, Kwok-Yan Lam, Liang Lin, & Keze Wang. Failure-Driven Workflow Refinement. ICML, 2026. (Spotlight)** ICML 2026
- Jusheng Zhang, Yijia Fan, Kaitong Cai, Jing Yang, Jiawei Yao, Jian Wang, Guanlong Qu, Ziliang Chen, & Keze Wang. Why Keep Your Doubts to Yourself? Trading Visual Uncertainties in Multi-Agent Bandit Systems. ICLR, 2026. (Accepted)** ICLR 2026
- Kaitong Cai, Jusheng Zhang, Yijia Fan, Jing Yang, & Keze Wang. RaCoT: Plug-and-Play Contrastive Example Generation Mechanism for Enhanced LLM Reasoning Reliability. AAAI, 2026. (Oral Presentation)** AAAI 2026
- Yijia Fan, Jusheng Zhang, Kaitong Cai, Jing Yang, Jian Wang, & Keze Wang. 3DAlign-DAER: Dynamic Attention Policy and Efficient Retrieval Strategy for Fine-grained 3D-Text Alignment at Scale. AAAI, 2026. (Accepted)** AAAI 2026
- Yijia Fan, Jusheng Zhang, Kaitong Cai, Jing Yang, & Keze Wang*. CCG: Rare-Label Prediction via Neural SEM-Driven Causal Game. EMNLP, 2025. (Accepted)** EMNLP 2025

Preprints & Projects

LoViF @CVPR26

Efficient Multimodal Rating Prediction with SmolVLM2

Mar. 2026

1st Place, Challenge on Efficient VLM for Multimodal Creative Quality Scoring

- Won **1st place** on the official leaderboard with CES = 0.43 and PLCC+ = 0.41.
- Built an efficient SmolVLM2-based multimodal regressor with mask-aware pooling and a lightweight MLP head.
- Optimized single-pass inference under efficiency constraints, using 152.12M parameters and 31.14 GFLOPs.

arXiv Preprint

Large Language Models as Discounted Bayesian Filters

arXiv:2512.18489

Jensen Zhang, Jing Yang, & Keze Wang.

- Proposed a **discounted Bayesian filtering framework** to characterize LLM in-context learning as online belief updating under non-stationary environments.
- Designed dynamic probabilistic probes, including **Biased Die** and **Gaussian Mean** tasks, and fit model-specific discount factors to quantify evidence forgetting.
- Showed that LLMs behave as **principled discounted updaters**, with errors mainly from prior/model misspecification, and linked belief-update quality to attention and hidden-state dynamics.

arXiv Preprint

Backward-Friendly Optimization: Training LLMs with Approximate Gradients under Memory Constraints

arXiv:2510.22467

Jing Yang, Kaitong Cai, Yijia Fan, Yufeng Wang, & Keze Wang.

- Proposed **GradLite**, a backward-friendly optimizer for memory-efficient full fine-tuning of LLMs with approximate gradients.
- Combined **low-rank Jacobian approximation** and **error-feedback correction** to reduce activation memory while preserving convergence guarantees.
- Achieved lower peak VRAM than checkpointed SFT (**38.7GB vs. 65.4GB**) with competitive performance on MMLU, GSM8K, multilingual, and MT-Bench evaluations.

HCI Project

LingoPal: Interactive HCI System for Post-Stroke Aphasia Rehabilitation

Personal Full-Stack

- Developed an end-to-end HCI rehabilitation system integrating wearable tongue-motion sensing hardware, Arduino-based multi-channel data acquisition, and a mobile interaction interface.
- Built a full-stack mobile application for real-time sensor data collection, speech/tongue-movement analysis, personalized training workflows, and visualized rehabilitation feedback.
- Served as the sole system designer and integrator, coordinating embedded sensing, data processing, UI interaction, and cross-module communication into a unified prototype.

Technical Skills

Programming: Python, PyTorch (Advanced), Golang, C++, MATLAB

Tools: Linux, Docker, Git, MindSpore, Arduino

Languages: English (IELTS: 7.5), Mandarin (Native)